

NATALIE DELLAMARIA

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EXPERIENCE

Distyl AI — New York, NY

Summer 2025

AI Platform Engineering Intern

- Built backend infrastructure and developer tooling for AI-native enterprise workflows, including validation frameworks, automated change summaries, and file-upload support for chat agents.
- Improved platform reliability by consolidating multi-step API workflows into a single atomic transaction, eliminating partial-failure states and simplifying recovery.

Tesla Energy — Palo Alto, CA

2020 – 2024

Staff Distributed Systems Engineer / Technical Lead

- Promoted from Engineer → Senior → Staff / Technical Lead in 4 years; led design and operation of backend cloud services supporting bidirectional communication with millions of IoT energy devices.
- Architected a high-throughput weather forecasting pipeline in Python, Kafka, and AWS serving 300K+ devices, enabling battery optimization across distributed energy systems.
- Designed and built APIs for Tesla's [Virtual Power Plant](#) platform supporting multiple secure dispatch modes and powering the customer mobile app experience.
- Led root-cause analysis of a production incident destabilizing the Kauai electrical grid, spanning multiple service layers, and presented corrective actions to senior leadership.
- Led GDPR readiness across distributed services and drove cross-functional delivery of [Charge on Solar](#) and [Tesla Electric Home Charging Plan](#) across firmware, cloud, security, and product.
- Built self-serve data pipelines that reduced recurring on-call and engineering support overhead; participated in 24/7 on-call for mission-critical energy services.
- Presented technical proposals to SVP-level leadership and delivered deep dives on VPP architecture to audiences of 500+ employees; mentored engineering interns.

EDUCATION

Harvard Business School — Cambridge, MA

2024 – 2026

MBA Candidate, Class of 2026

AI Club Co-President; Technical AI Safety Fellow, AI Student Safety Team.

Research: Demand response potential from data centers, including load flexibility and energy market participation.

Clemson University — Clemson, SC

2016 – 2020

B.S. Computer Science, Summa Cum Laude • Clemson University National Scholar

TECHNICAL SKILLS

Languages and Frameworks: Python, Scala, Akka

Infrastructure: Kafka, Kubernetes, Docker, AWS, PostgreSQL, Redis

Distributed Systems: Event-driven architectures, microservices, gRPC, streaming pipelines, actor models

Domains: Real-time data processing, IoT infrastructure, cloud-native systems, distributed energy systems

PROJECTS & PRESENTATIONS

Analog Image Classifier — Transfer-learning model (ResNet50) for analog film photographs; built and labeled custom dataset, fine-tuned in TensorFlow, tracked experiments in Weights & Biases.

QCon San Francisco (2022) — [Tesla's Virtual Power Plant](#).

HBS AI Club (2024) — Decoding Deepfakes: The Role of GANs in Synthetic Video Creation.